

OVERVIEW

- BROS2 is an Electron desktop IDE that lets you assemble ROS2 node graphs as draggable blocks and auto-generates packages, launch files, and a Docker-backed workspace. BROS2 will be known as “Scratch for ROS2,” as it enables rapid iteration on robot behaviors with live telemetry and validation, without writing extensive code.

MOTIVATION

- ROS 2 is difficult to work with and implement: you must understand packages, launch files, build tools, and Docker before you can test even simple behaviors. BROS2 lowers this barrier so students and teams can prototype multi-node systems quickly and reproducibly.

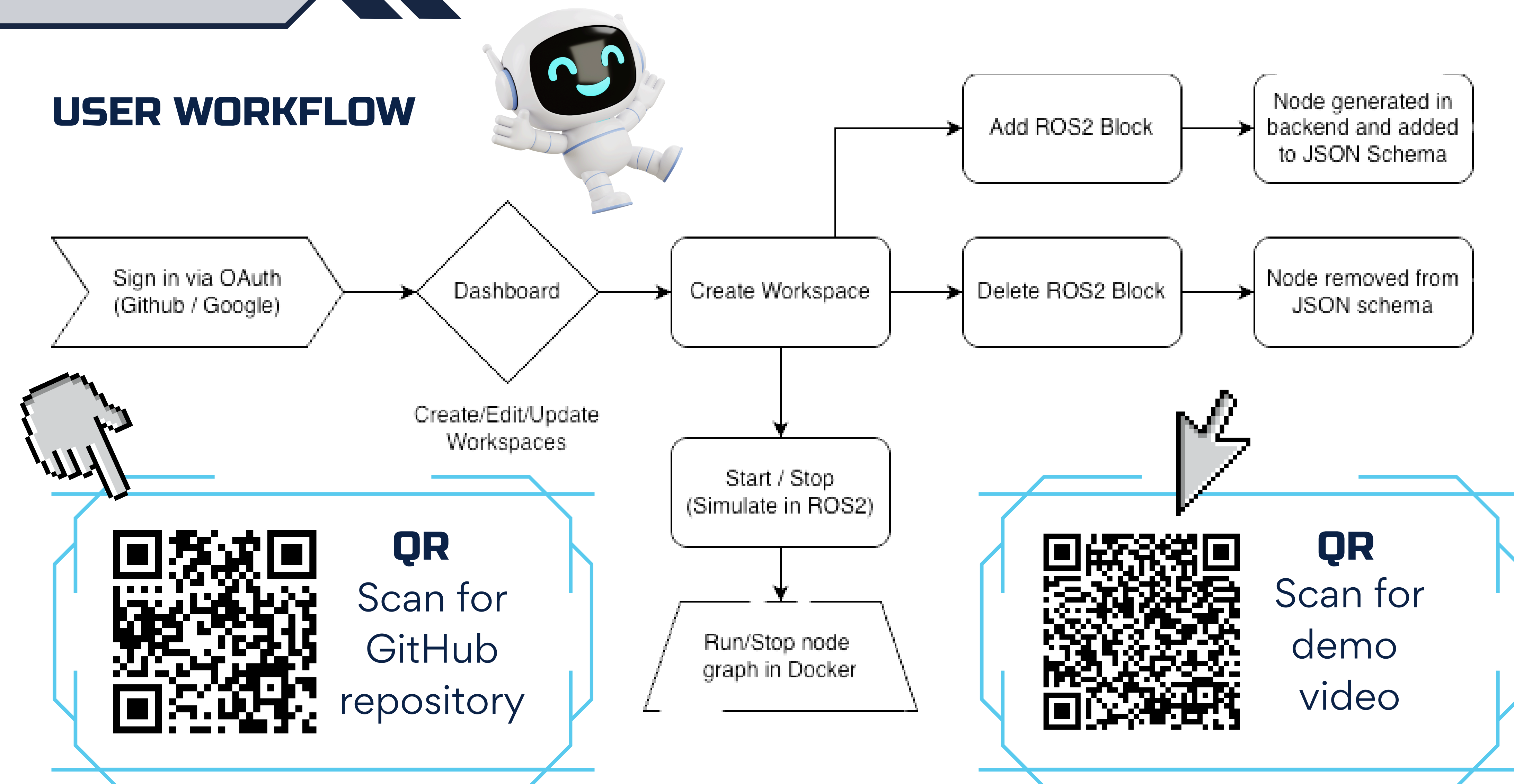
SYSTEM ARCHITECTURE

- The React canvas stores a block graph describing nodes, topics, and parameters. A TypeScript IR layer validates the graph and reports issues in real time. When a workspace is run, a preload bridge talks to a Docker “runner” service that builds and launches ROS 2 inside a container.

NEXT STEPS

- Next steps: auto-emitting full ROS 2 packages, richer block libraries, and integration with Gazebo/Isaac simulation.

USER WORKFLOW



BACKEND ARCHITECTURE

